Rank-two geometries from primitive residues

Human Compression project

Working proof, 14 October 2026

Let G be a finite group. For proper subgroups $A, B < G$, put $H = A \cap B$. The coset incidence structure $\Gamma(G; A, B)$ has the right cosets of A and of B as its two types. Two cosets are incident when they intersect. We allow an isomorphism induced by an automorphism of G to interchange the two types. This is the equivalence used throughout this note.

We consider connected, firm, flag-transitive geometries for which at least one of the two actions on a type is primitive and both actions on the neighbours of a vertex are doubly transitive. In rank two these are the weak primitivity and local double transitivity conditions. The intersection property requires a partial linear space or a complete bipartite geometry. All group actions below are the specified actions of G; we do not replace them by the full automorphism group of the incidence graph.

Theorem 1. *Every geometry with these properties is represented by the following finite construction.*

1. *Choose A from a complete set of conjugacy classes of maximal proper subgroups of G.*
2. *Choose H from a complete set of A-conjugacy classes of maximal proper subgroups of A, retaining those for which A acts doubly transitively on A/H.*
3. *Choose each actual subgroup B with $H < B < G$. Retain it when $A \cap B = H$ and the action of B on B/H is doubly transitive.*
4. *Apply the incidence test in Lemma 2, and then identify pairs under $\mathrm{Aut}(G)$ and interchange of the two entries.*

Conversely, every retained pair defines such a geometry.

Proof. Choose a flag of an arbitrary geometry and let A, B be its vertex stabilizers. Flag transitivity identifies its incidence relation with intersection of the corresponding cosets. Primitivity on one type says that its vertex stabilizer is maximal; interchange the types if necessary. Conjugating the flag puts A in the chosen list. Its flag stabilizer is $H = A \cap B$. Double transitivity on the neighbours implies primitivity on them, so H is maximal in A. An additional conjugation within A puts H in the second list without changing A. The conjugated B occurs among the actual overgroups in the third step. Thus none of the required geometries is omitted.

Conversely, both indices $[A : H]$ and $[B : H]$ are at least two, giving firmness. The strict inclusion $H < B$ and the equality $A \cap B = H$ imply $B \not\subseteq A$. Maximality of A gives $\langle A, B \rangle = G$, which is exactly connectivity of the coset incidence graph. Right multiplication is transitive on flags: an incident pair has a common element g and is the translate (Ag, Bg) of the base flag. The two local actions are precisely those on A/H and B/H, and the action on the cosets of A is primitive. The incidence test supplies the remaining intersection condition. Finally, an automorphism of G taking one subgroup pair to another induces the stated equivalence of coset geometries. This last quotient neither discards an orbit nor changes the equivalence convention. □

Lemma 2. *Assume the conditions of the first three steps. Choose $a \in A \setminus H$ and a right transversal T of H in B. Define*

$$c = \big|\{b \in T : \text{ some } d \in T \text{ satisfies } b(da)^{-1} \in A\}\big|.$$

The geometry has the intersection property if and only if $c = 1$ or $[B : H] = [G : A]$.

Proof. The two distinct neighbours B and Ba of A have respective neighbour sets $\{Ab : b \in T\}$ and $\{Ada : d \in T\}$. Entries in each set are distinct, because $A \cap B = H$. Equality of two such right cosets is equivalent to $b(da)^{-1} \in A$, so their number of common neighbours is exactly c. Double transitivity of A makes this number the same for every pair of distinct neighbours of A. Transitivity on vertices of that type makes it the same for every pair having a common neighbour. Consequently $c = 1$ is equivalent to the absence of a quadrangle, hence to the partial linear space condition. The graph is complete bipartite exactly when each vertex of the second type is adjacent to all $[G : A]$ vertices of the first type. Its degree is $[B : H]$, proving the other alternative. These exhaust the two permitted intersection conditions. □

Application to the Higman–Sims group

The current computation uses the degree 100 permutation copy of HS already fixed in the project's reviewed higher-rank roster. It does not assume that roster is complete. The subgroup classifications supplied by GAP's table of marks library are retained inputs. In particular, the file `tmhsmcl.tom`, credited to Thomas Merkwitz and present in its recorded history in 1997, predates the 2005 article considered here. A library lookup and an isomorphism to its underlying group are not a fresh proof of that subgroup classification.

There are twelve classes of maximal subgroups. The local test leaves 38 choices of (A, H); their actual overgroups give 13 pairs satisfying both double transitivity conditions. Three fail the incidence test: one has $c = 6$ and two have $c = 90$. The ten remaining pairs form six orbits under $\mathrm{Aut}(HS)$ with type interchange. Their orders and local degrees are:

$\lvert A\rvert$	$\lvert B\rvert$	$\lvert H\rvert$	$[A : H]$	$[B : H]$
443520	40320	20160	22	2
10752	20160	1344	8	15
10752	7680	1536	7	5
7920	7920	660	12	12
1200	3840	240	5	16
1200	1000	200	6	5

Different rows are inequivalent already by the unordered pair of parabolic orders. The four additional pairs are identified with their row representatives by explicit automorphisms, recorded as images of generators of G. Each map is checked to be a bijective homomorphism and to carry both parabolics to their targets, with the stated interchange of types. Thus this quotient does not need a separate completeness assertion for a computed automorphism group. The search uses the previously reviewed simultaneous subgroup matcher. The full outputs, including the three rejected pairs, are retained beside this working manuscript.

This computation reproduces the rank-two count in D. Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics **33** (2005), 231–236, Section 2. The general construction and its proof above separate the finite local actions, actual overgroups and incidence test; the classical coset correspondence and subgroup classifications are borrowed. No claim of historical novelty for the enumeration principle is made.